Kural: Frontend Requirements and API

For the frontend owner | React + Vite + Tailwind (JavaScript) | 12-hour build | Feature freeze at hour 8

Kural turns a spoken complaint (Tamil or English) plus a photo into the right government document. The backend does all AI, routing and PDF work. The frontend's job is to make that flow fast and obvious for an elderly citizen or a kiosk operator: record, add photos, answer any questions, review the draft, download it, copy the portal fields, and hear the next steps.

1. Scope

#	Frontend feature	Priority	Done when
F1	Voice recorder	P0	Records up to 30 s per clip with a visible countdown and auto-stop; user can add a second clip; transcript shown in an editable box
F2	Typed input fallback	P0	User can type or edit the complaint text instead of recording
F3	Photo upload	P0	0-3 photos from camera or gallery, thumbnails, remove button
F4	Follow-up questions	P0	Shows backend questions in the selected language, sends answers, loops until ready (max 2 rounds)
F5	Photo mismatch alert	P0	Explains the mismatch; user can retake photos or continue anyway
F6	Route display	P0	Shows route, rule id and reason; operator can override the route
F7	Complainant details	P0	Name, address, phone form before the draft is generated
F8	Draft preview and edit	P0	Rendered draft visible; editable fields saved via PATCH
F9	Portal fields	P0	Each CM Helpline field with a one-tap copy button and 'Copied' feedback
F10	PDF download	P0	One tap downloads the bilingual PDF
F11	Next steps	P0	Text in both languages plus a Play button using /api/tts; text only if TTS is unavailable
F12	Mark as filed	P0	Saves the complaint to history (POST /api/complaints)
F13	UI language toggle	P0	Whole UI switches Tamil / English
F14	My complaints tracker	P1	List with filing date, days left, status badges
F15	One-tap escalation	P1	Escalate button on overdue complaints opens the new draft
F16	Online / offline toggle	P1	Header switch with model status from /api/health
F17	Use my location	P1	Sends browser lat/lon with the complaint for ward detection

Out of scope: login, accounts, submitting to government portals, languages beyond Tamil and English, a native app.

2. Setup and conventions

- **Stack:** React 18 + Vite + Tailwind CSS, plain JavaScript (.jsx). No TypeScript, no state library; React state and context are enough.
- **API base:** every call goes to relative `/api/...`. In `vite.config.js`, proxy `/api` to `http://localhost:8000`.
- **One API module:** all requests live in `src/api/client.js`, one function per endpoint. Components never call fetch directly.
- **Mocks first:** with `VITE_USE_MOCKS=true`, client.js returns the example responses in section 6 after a 1-second delay. Build the whole UI against mocks; switch off when the backend is ready.
- **Works offline:** the kiosk demo runs with WiFi off. Bundle everything locally: no CDN scripts, no Google Fonts links. Put Noto Sans Tamil font files in `src/assets/fonts`.
- **Microphone needs HTTPS** on phones. Locally, localhost is fine; for judges' phones use the Cloudflare Tunnel HTTPS link.

```
// vite.config.js
export default defineConfig({
  plugins: [react()],
  server: { proxy: { '/api': 'http://localhost:8000' } },
})
```

3. Screens and user flow

Single-page app with two routes: **New complaint** (a step-by-step wizard) and **My complaints** (P1). The header holds the app name, the UI language toggle and, in P1, the online/offline switch.

Step	Screen	What the user sees and does	API calls
1	Describe	Big record button with 30 s countdown, 'Add another clip', editable transcript box, photo upload (0-3), optional 'Use my location' (P1), Continue button	POST /transcribe (per clip), then POST /sessions on Continue
2	Clarify	Shown only when status is not ready. needs_info: question cards with input boxes. photo_mismatch: alert with Retake photos / Continue anyway. not_civic: message and Start over	POST /sessions/{id}/answers
3	Route and details	Route card (route name, rule id, reason in UI language). Override dropdown for operators. Complainant form: name, address, phone. Generate draft button	POST /drafts
4	Draft	Draft preview, Edit mode for editable fields, portal fields with copy buttons, next steps with Play button, Download PDF, Mark as filed	PATCH /drafts/{id}, GET /drafts/{id}/pdf, POST /tts, POST /complaints
5	Done	Confirmation with due date and 'File another complaint'	-
P1	My complaints	Cards with title, route, filed date, days left, status badge; Escalate on overdue items	GET /complaints, POST /complaints/{id}/escalate

Session status handling

status from backend	Frontend behaviour
needs_info	Go to Clarify, render follow_up_questions (question_ta or question_en by UI language). Submit answers. After 2 rounds the backend stops asking; continue with what is filled.
photo_mismatch	Show extraction.photo_mismatch_note. 'Retake photos' returns to Describe with text kept and photos cleared (a new session is created). 'Continue anyway' sends confirm_photo_mismatch: true.
not_civic	Show: 'Kural can only help with civic problems like roads, streetlights, garbage, water and drains.' Offer Start over.
ready	Go to Route and details. decision is always present when ready.

Display names for routes

route	English label	Tamil label key
local_body	Local body complaint	route.local_body
cm_helpline	CM Helpline grievance	route.cm_helpline
rti	RTI application	route.rti

4. Components

Component	Responsibility	Key props / state
useRecorder (hook)	MediaRecorder wrapper; 30 s auto-stop; returns blob	start(), stop(), isRecording, secondsLeft, blob, error
Recorder.jsx	Record button, countdown ring, clip list, calls transcribe	language, onTranscript(text)

Component	Responsibility	Key props / state
PhotoUpload.jsx	accept='image/*' capture='environment', max 3, thumbnails	photos, onChange(files)
FollowUpQuestions.jsx	One input per question, submit all	questions, uiLang, onSubmit(answers)
PhotoMismatchAlert.jsx	Mismatch note + two actions	note, onRetake(), onContinue()
RouteBadge.jsx	Route label, rule id chip, reason	decision, uiLang, onOverride(route)
ComplainantForm.jsx	Name (required), address (required), phone	value, onChange
DraftPreview.jsx	Renders draft.html in a sandboxed iframe (srcDoc); edit mode for editable fields	draft, onSave(fields)
PortalFields.jsx	Label + value + Copy button per field	fields (object)
NextSteps.jsx	Both languages, Play button, falls back to text on 501	nextSteps, uiLang
ModeToggle.jsx (P1)	Online/offline switch, status dot	reads /health, calls PUT /config/mode
ComplaintCard.jsx (P1)	Tracker row with status and Escalate	complaint, onEscalate(id)

Editable draft fields (send only these in PATCH `fields`): `summary_en, summary_ta, location_text, duration, requested_info`. Legal wording is not editable.

5. UX requirements

- **Mobile-first, kiosk-friendly:** large touch targets (min 48 px), base font 18 px, high contrast, one main action per screen.
- **Loading states with honest waits:** model calls can take 5-15 s online and longer offline. Show a spinner with a step label ('Understanding your complaint...'), disable buttons while waiting, and never block without feedback.
- **Transcript is editable** before Continue, so speech-to-text mistakes can be fixed.
- **Countdown is always visible** while recording; recording stops automatically at 0.
- **Copy buttons** use navigator.clipboard.writeText with a 'Copied' state for 2 s.
- **PDF download** uses an anchor to /api/drafts/{id}/pdf with the download attribute; no new tab.
- **Privacy:** do not store complaint text, photos or personal details in localStorage. Keep the current session_id in sessionStorage only so a refresh can recover via GET /sessions/{id}.
- **Language:** UI language (toggle) is separate from complaint language (sent to the backend). Default both to Tamil. All UI strings come from src/i18n/ta.json and en.json; no hard-coded text.
- **Errors** appear as a clear banner with a retry action where it makes sense (section 7).

6. Backend endpoints

All paths are under `/api`. JSON unless stated. These are the agreed contracts; if the backend changes one, update this page and client.js together.

Method	Path	Used on	Priority
GET	/api/health	Header status, app start	P0
POST	/api/transcribe	Describe (each clip)	P0
POST	/api/sessions	Describe -> Continue	P0
POST	/api/sessions/{id}/answers	Clarify	P0
GET	/api/sessions/{id}	Recover after refresh	P0
POST	/api/drafts	Route and details -> Generate	P0
PATCH	/api/drafts/{id}	Draft edit -> Save	P0
GET	/api/drafts/{id}/pdf	Download PDF	P0
POST	/api/tts	Next steps -> Play	P0
POST	/api/complaints	Mark as filed	P0
GET	/api/complaints	My complaints	P1
POST	/api/complaints/{id}/escalate	Escalate button	P1
PUT	/api/config/mode	Online/offline toggle	P1

GET /api/health

Call on app start and every 30 s while the header is visible.

Response

```
{ "status": "ok", "model_mode": "online", "model": "gemma-4" }
```

POST /api/transcribe

multipart/form-data. Send one clip per call.

Request

```
audio:    <Blob webm or wav>        (from MediaRecorder)
language: "ta" | "en"
```

Response

```
{ "transcript": "...", "language": "ta", "engine": "gemma" }
```

Append each transcript to the editable text box, separated by a space.

POST /api/sessions

multipart/form-data. Starts analysis.

Request

```
text:     "full complaint text"     (required)
language: "ta" | "en"
photos:   File (0-3, repeat the field for each photo)
lat, lon: numbers                   (optional, P1)
```

Response

```
{
  "session_id": "s_8f2a",
  "status": "needs_info",          // needs_info | photo_mismatch | not_civic | ready
  "extraction": {
    "category": "streetlight",
    "summary_en": "The streetlight ... has not worked for 3 months despite 2 complaints.",
    "summary_ta": "...",
    "location_text": null,
    "duration": "3 months",
    "severity": "medium",
    "photo_findings": ["Unlit streetlight pole"],
    "photo_matches_claim": true,
    "photo_mismatch_note": null,
    "wants_records": false,
    "user_says_ignored": true,
    "requested_info": [],
    "missing_fields": ["location_text"]
  },
  "decision": {
    "route": "cm_helpline", "rule_id": "R3",
    "reason_en": "Your earlier complaints were ignored, so this goes to the CM Helpline.",
    "reason_ta": "...", "escalation_level": 1
  },
  "ward": null,
  "follow_up_questions": [
    { "field": "location_text",
      "question_en": "Which street is this on?", "question_ta": "..." }
  ]
}
```

decision may be null until status is ready.

POST /api/sessions/{id}/answers

Answer questions, or confirm a photo mismatch.

Request

```
{ "answers": { "location_text": "Gandhi Street, near the bus stand" },
  "confirm_photo_mismatch": false }
```

Response

```
Same shape as POST /api/sessions.
```

GET /api/sessions/{id}

Use only to recover after a page refresh.

Response

```
Same shape as POST /api/sessions.
```

POST /api/drafts

Only when status is ready; otherwise the backend returns 409 SESSION_NOT_READY.

Request

```
{ "session_id": "s_8f2a",
  "complainant": { "name": "R. Priya", "address": "14 Gandhi Street", "phone": "98xxxxxx10" },
  "route_override": null }        // or "local_body" | "cm_helpline" | "rti"
```

Response

```
{
  "draft_id": "d_31c9",
  "decision": { "route": "cm_helpline", "rule_id": "R3", "escalation_level": 1 },
  "department": { "id": "muni-electrical", "name": "...", "address": "...",
                  "submit_via": "...", "response_days": 30 },
  "html": "<html>...bilingual rendered template...</html>",
  "portal_fields": { "Grievance details": "...", "Communication address": "..." },
  "next_steps": { "en": "Submit at ... reply within 30 days.", "ta": "..." }
}
```

PATCH /api/drafts/{id}

Send only changed fields.

Request

```
{ "fields": { "summary_en": "edited text" }, "complainant": { "phone": "..." } }
```

Response

```
Same shape as POST /api/drafts (re-rendered html and portal_fields).
```

GET /api/drafts/{id}/pdf

Returns application/pdf, filename kural_<route>_<draft_id>.pdf. Use a plain download link.

POST /api/tts

Returns audio/wav. Play with new Audio(URL.createObjectURL(blob)).

Request

```
{ "text": "...", "language": "ta" }
```

501 means no offline TTS engine is available: hide the Play button and show text only. Cache the audio per language so repeat plays don't refetch.

POST /api/complaints

Called by Mark as filed.

Request

```
{ "draft_id": "d_31c9", "filed_on": "2026-10-26" }
```

Response

```
{ "id": "c_7a1", "draft_id": "d_31c9", "route": "cm_helpline", "category": "streetlight",
  "location_text": "Gandhi Street", "ward": null,
  "filed_on": "2026-10-26", "due_on": "2026-11-25",
  "status": "filed", "parent_complaint_id": null }
```

GET /api/complaints (P1)

Response

```
[ { "id": "c_7a1", "draft_id": "d_31c9", "title": "Streetlight, Gandhi Street",
    "filed_on": "2026-10-26", "due_on": "2026-11-25",
    "days_left": 30, "status": "filed" } ]      // filed | overdue | resolved | escalated
```

Sort overdue first, then by days_left ascending. Escalate button only on overdue.

POST /api/complaints/{id}/escalate (P1)

No body. Returns a new draft (same shape as POST /api/drafts); open it on the Draft screen.

PUT /api/config/mode (P1)

Request

```
{ "mode": "offline" }
```

Response

```
Same as /api/health. 503 means Ollama is not running: revert the switch and show the error.
```

7. Error handling

Every error uses one shape. Show `message` only for debugging; show the friendly text below to users (both languages in i18n).

```
{ "error": { "code": "MODEL_INVALID_OUTPUT", "message": "..." } }
```

HTTP	code	What the user sees	Action
400	BAD_INPUT	Please check your complaint text and photos (max 3).	Stay on screen
404	NOT_FOUND	This complaint was not found. Please start again.	Start over
409	SESSION_NOT_READY	We still need a few details.	Go to Clarify
422	MODEL_INVALID_OUTPUT	We couldn't understand that. Try again or type it.	Retry button
422	NO_DEPARTMENT	We couldn't find the right office. The operator can choose one.	Show route override
501	(tts only)	-	Hide Play button
503	MODEL_UNAVAILABLE	The AI service is unavailable. Check the connection or switch to offline mode.	Retry
504	MODEL_TIMEOUT	This is taking too long. Please try again.	Retry
-	network error	Can't reach Kural. Is the server running?	Retry

8. Language strings

Both files share the same keys. Minimum keys to create:

```
app.title, app.tagline, lang.toggle,
step.describe, step.clarify, step.route, step.draft, step.done,
rec.start, rec.stop, rec.secondsLeft, rec.addClip, rec.micDenied,
text.placeholder, photos.add, photos.max, location.use,
btn.continue, btn.back, btn.retry, btn.startOver, btn.generate,
btn.edit, btn.save, btn.download, btn.copy, btn.copied, btn.play, btn.markFiled, btn.escalate,
clarify.title, mismatch.title, mismatch.retake, mismatch.continue, notCivic.message,
route.title, route.rule, route.override, route.local_body, route.cm_helpline, route.rti,
form.name, form.address, form.phone, form.required,
draft.title, portal.title, nextSteps.title, done.title, done.dueOn,
complaints.title, complaints.daysLeft, status.filed, status.overdue, status.resolved, status.escalated,
loading.understanding, loading.drafting, loading.transcribing,
error.* (one per code in section 7), mode.online, mode.offline
```

9. Frontend build plan (12 hours)

Hours	Work	Check
0-1	Vite + Tailwind setup, proxy, router, header with language toggle, i18n files, client.js with mocks	App runs with mock data
1-3	useRecorder + Recorder with countdown, PhotoUpload, Describe screen	Record 30 s, auto-stop, transcript editable
3-4	Clarify screen: questions, mismatch alert, not_civic	All 4 statuses work on mocks
4-6	Route and details, Draft screen: preview, edit, portal fields, PDF, next steps, mark filed	Full flow on mocks
6-7	Switch to real backend, fix contract mismatches, error banners	Full flow on real API

Hours	Work	Check
7-8	P1: tracker, escalation, mode toggle, location	Only after P0 works
8-12	Feature freeze: bugs, phone testing over HTTPS, two full demo rehearsals with WiFi off	Demo never breaks

Demo checklist

- Tamil UI loads with correct Tamil font and no internet connection.
- Microphone permission works on the judges' phones via the HTTPS tunnel link.
- Prepared demo: streetlight complaint with photo shows a follow-up question, then CM Helpline route R3, draft, copy fields, PDF and spoken next steps.
- Photo mismatch demo: say 'pothole', upload a garbage photo, the alert appears.
- Offline mode: switch to offline, repeat the flow with WiFi off.